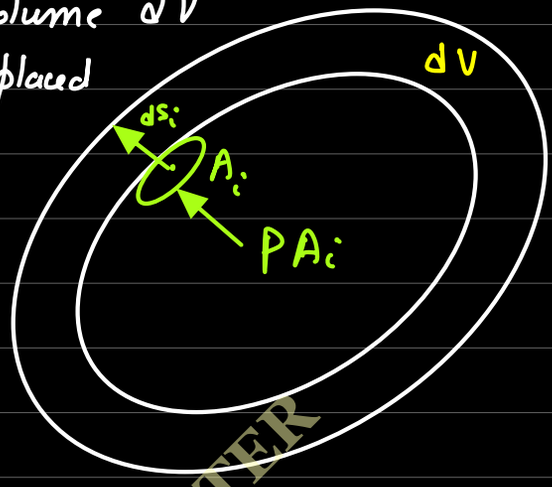
Work done by a Gas: Recall that work done by a force is given as

$dW = \vec{F} \cdot d\vec{S}$, Now Consider a confined gas having a volume V and a pressure P expanding by a volume dV



Now Consider a patch of area A_i being displaced through a displacement dS_i. Then total work done on all such patches is given as $dW_i = PA_i\,dS_i = P\,dV_i$

Where dV_i is the volume swept by the patch A_i. Thus total work done by the gas is simply:

$$dW = \sum dW_i = P \sum dV_i = P\,dV.$$

Polytropic Process: A process of the form $PV^n = \text{Constant}$ is known as a polytropic process. This process is of special interest to us because by varying n, we can generate various standard processes, e.g:

$n = 0 \Rightarrow P = \text{Constt}$ (Isobaric Process).

$n = 1 \Rightarrow PV = \text{Constt} = T = \text{Constt}$ (Isothermal Process)

$n = \infty \Rightarrow V P^{1/\infty} = \text{Constt} \Rightarrow V = \text{Constt}$ (Isochoric Process)

(Note $PV^n = \text{Constt} \equiv VP^{1/n} = \text{Constt}$).

$n = \gamma \Rightarrow$ Adiabatic Process (Will be proved later).

$n = -1 \Rightarrow$ Constant collision frequency process.

Thus analysing a general polytropic process will give us insights in to all the standard processes.

Note that polytropic process can be alternately described by any of the following equations:

$$\boxed{PV^n = \text{Constt} \equiv TV^{n-1} = \text{Constt} \equiv PT^{\frac{n}{1-n}} = \text{Constt}.}$$

This can be shown as follows: $PV^n = \text{Constt}$, $PV = RT$

$$\Rightarrow V = \frac{RT}{P} \Rightarrow P\left(\frac{RT}{P}\right)^n = \text{Constt} \Rightarrow P^{1-n} T^n = \text{Constt}$$

$$\Rightarrow PT^{\frac{n}{1-n}} = \text{Constt}. \quad \text{Also } P = \frac{RT}{V} \Rightarrow \left(\frac{RT}{V}\right) V^n = \text{Constt} \Rightarrow TV^{n-1} = \text{Constt}.$$

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

Example: Find the polytropic coefficient x for a process in which collision frequency is constant.

Solution: Recall from XI G that $z = \sqrt{2}\,\eta\,\pi D^2 \bar{v}$ where η is particle density. Now $\eta \propto \frac{1}{V}$ and $\bar{v} \propto \sqrt{T}$

$\Rightarrow$ For constant collision frequency, $\frac{\sqrt{T}}{V}$ = Constant $\Rightarrow T\bar{V}^2$ = Constt.

Comparing with polytropic form TV^{x-1} = Constt, $x = -1$. Ans.

Example: In a certain process, when pressure of a gas was doubled, collisions per unit area per unit time with vessel walls became 16 times, how many times did the temperature change? What is the polytropic coefficient? (Assuming polytropic process).

Solution: $\frac{P_2}{P_1} = 2$ (1) $\frac{z_2}{z_1} = 16$, Recall $z = \frac{1}{4}\eta\bar{v}$

Now $\eta \propto \frac{1}{V}$, $\bar{v} \propto \sqrt{T}$ $\Rightarrow \left(\frac{\sqrt{T_2}}{V_2}\right) \Big/ \left(\frac{\sqrt{T_1}}{V_1}\right) = 16$

$\Rightarrow \frac{V_1}{V_2}\sqrt{\frac{T_2}{T_1}} = 16$ (2)

(1)/(2) $\Rightarrow \frac{P_2 V_2}{P_1 V_1}\sqrt{\frac{T_1}{T_2}} = \frac{1}{8} \Rightarrow \sqrt{\frac{T_2}{T_1}} = \frac{1}{8} \Rightarrow T_2 = \frac{T_1}{64}$

$\Rightarrow T_2 = \frac{T_1}{64}$ Ans.

Now $\frac{P_2}{P_1} = 2$, $\frac{T_2}{T_1} = \frac{1}{64}$ $\Rightarrow \left(\frac{P_2}{P_1}\right)^6 \times \left(\frac{T_2}{T_1}\right) = 1$

$\Rightarrow P_2 T_2^{1/6} = P_1 T_1^{1/6}$ = Constt.

Now $P T^{\frac{x}{1-x}}$ = Constt $\Rightarrow \frac{x}{1-x} = \frac{1}{6} \Rightarrow x = \frac{1}{7}$ Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER

Molar Specific Work for a Polytropic Process. (Ω): For a polytropic process we shall call the work done per unit rise in temperature per mol of gas as molar specific work (Ω), mathematically,

$$\boxed{\Omega = \frac{1}{n}\frac{dW}{dT}}$$

. It turns out that molar specific work for any polytropic process comes out to be constant (independent of temperature),

$$\boxed{\Omega = \frac{R}{1-x}}$$

Now if we have n moles of gas heated through a temperature ΔT, then, $\boxed{\Delta W = n\Omega\Delta T}$

Proof of molar specific work equation for polytropic process:

$PV = nRT$, $PV^x = \text{Constt} \Rightarrow TV^{x-1} = \text{Constt.}$

$$\Rightarrow \ln T + (x-1)\ln V = \text{Constt} \Rightarrow \frac{1}{T} + \left(\frac{x-1}{V}\right)\frac{dV}{dT} = 0$$

$$\Rightarrow \frac{dV}{dT} = \frac{V}{T(1-x)} \Rightarrow \frac{PdV}{dT} = \frac{PV}{T(1-x)} = \frac{dW}{dT}$$

$$\Rightarrow \frac{dW}{dT} = \frac{nR}{1-x} \Rightarrow \frac{1}{n}\frac{dW}{dT} = \frac{R}{1-x}$$

$$\Rightarrow \boxed{\Omega = \frac{R}{1-x}}$$

Example: 2 mols of an ideal gas undergo a constant collision frequency process in which the temperature was doubled from T_0 to $2T_0$. Find the work done in the process. Also find how many times did pressure and volume changed in the process.

Solution: $\Delta W = n\Omega\Delta T$, $x = -1$ (From a previous example).

YOUTUBE.COM/@THEPHYSICSSHELTER

$\Rightarrow C = R/1-x = R/2 \Rightarrow \Delta W = 2 \times \frac{R}{2} \times (2T_0 - T_0)$

$\Rightarrow \Delta W = RT_0$ Ans. Now $PV^{-1} = C \Rightarrow TV^{-2} = \text{Constt.}$

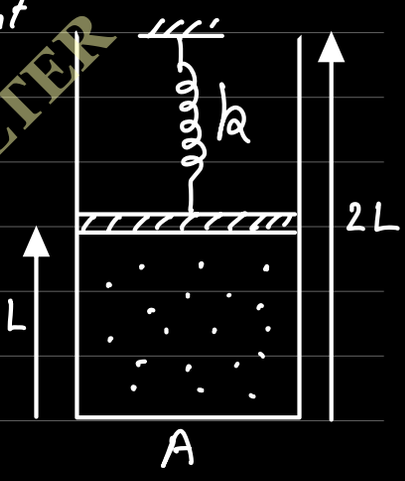
$\Rightarrow V \propto \sqrt{T}$, $\because$ T got doubled $\Rightarrow$ V became $\sqrt{2}$ times.

Also $PV^{-1} = \text{Constt} \Rightarrow P \propto V \Rightarrow$ Pressure also became $\sqrt{2}$ times.

Example: A cylinder contains a pressurised gas under a massless piston connected to spring of constant k and a natural length 2L. Surrounding the vessel is vacuum. Now the gas is expanded by supplying some heat.

(a) Is the proun polytropic?

(b) What is the work done by gas in increasing its volume 1.5 times?



Sol: If piston is at a height y, $P = \frac{ky}{A}$

$\Rightarrow \frac{P}{Ay} = \frac{k}{A^2} \Rightarrow PV^{-1} = \frac{k}{A^2} \Rightarrow n = -1.$ (! Polytropic).

$\Rightarrow C = \frac{R}{1-n} = R/2. \Rightarrow \Delta W = n C \Delta T = \frac{nR}{2} \Delta T$

Now $\frac{nR}{2}\Delta T = \frac{1}{2}\Delta(PV)$, $P_f = \frac{3}{2}\frac{kL}{A}$ $V_f = \frac{3}{2}AL$, $P_i = \frac{kL}{A}$, $V_i = AL$

$\Rightarrow \Delta W = \frac{1}{2}\left(\frac{9}{4} - 1\right) kL^2 = \frac{5}{8} kL^2$ Ans.

Note we could also find work by direct integration: $dW = PdV = \frac{ky}{A} \times A\,dy.$

$\Rightarrow \Delta W = k \int_{L}^{3L/2} y\,dy = \frac{5}{8} kL^2.$

YOUTUBE.COM/@THEPHYSICSSHELTER

First Law of Thermodynamics: Heat supplied to any system is equal to the sum of work done by the system and change in its internal energy. Mathematically:

$$dQ = dU + dW$$

Also: $\Delta Q = \Delta U + \Delta W$

$\Delta Q \rightarrow$ Heat supplied to the system, $\Delta W \rightarrow$ Work done by the system
$\Delta U \rightarrow$ Change in internal energy of the system.

→ First law of thermodynamics is essentially a statement of law of conservation of Energy and can be applied to all kinds of systems including mechanical, electrodynamic, nuclear, gravitational systems (for nuclear systems, energy equivalent of mass has to be considered)

→ While applying first law, it should be absolutely clear in mind as what exactly is the system being considered. (For instance in a setup with gas, pistons and springs we could possibly consider gas alone, or gas + spring or gas + piston as systems).

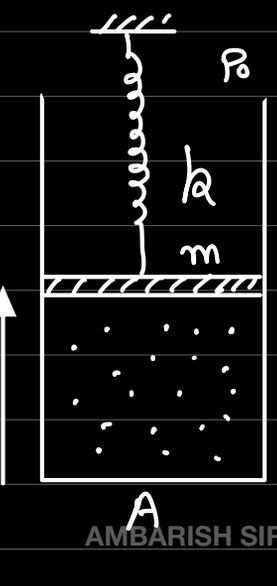
→ U includes all kinds of internal energies of the system including kinetic energy, molecular potential energy, gravitational potential energy for pistons, spring potential energies for springs.

Example: A gas is trapped under a massive piston in equilibrium with spring initially at natural length, as shown in the figure. Now the gas is heated until piston rises through a distance L.

(a) Is the process polytropic?

(b) Considering spring + piston + gas as system calculate ΔQ, ΔU and ΔW for the system by the time system rises through a distance L.

Atmospheric pressure is p_0. Degrees of freedom = f.



YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

Solution: Let the piston be at a height y from the bottom then:

$$P = p_0 + \frac{mg}{A} + \frac{k(y-L)}{A}, \quad V = Ay \Rightarrow y = \frac{V}{A}$$

$$\Rightarrow P = p_0 + \frac{mg}{A} - \frac{kL}{A} + \frac{kV}{A^2}$$

This is not polytropic in general unless $p_0 + \frac{mg}{L} - \frac{kL}{A} = 0$.

Now $U = n\frac{f}{2}RT = \frac{f}{2}PV \Rightarrow \Delta U = \frac{f}{2}\Delta(PV)$.

$$P_i V_i = \left(p_0 + \frac{mg}{A}\right)(AL), \quad P_f V_f = \left(p_0 + \frac{mg}{A} + \frac{kL}{A}\right)(2AL)$$

$$\Rightarrow \Delta U_{gas} = \left(mgL + 2kL^2 + p_0 AL\right)\frac{f}{2}$$

$$\Delta U_{piston} = mgL, \quad \Delta U_{spring} = \frac{1}{2}kL^2$$

$$\Rightarrow \Delta U = \Delta U_{gas} + \Delta U_{piston} + \Delta U_{spring} = mgL + \frac{1}{2}kL^2 + \frac{f}{2}\left(mgL + 2kL^2 + p_0 AL\right)$$

Now the system is pushing against a constant pressure atmosphere.
$\therefore \Delta W_{sys} = p_0 AL$

Now $\Delta Q_{sys} = \Delta U_{sys} + \Delta W_{sys} = p_0 AL + mgL + \frac{1}{2}kL^2 + \frac{f}{2}\left(mgL + 2kL^2 + p_0 AL\right)$ Ans.

Q. In the previous situation find the individual words by the spring, piston and the gas. (Assume piston has negligible heat capacity).

Solution: Spring is doing negative work on piston and negative of work done by spring force is its change in potential energy.

$$\Rightarrow \Delta W_{spring} = -\frac{1}{2}kL^2$$

Now Gas is pushing the piston which in turn is pushing the atmosphere and spring and gaining gravitational PE.

AMBARISH SIR

https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

YOUTUBE.COM/@THEPHYSICSSHELTER

Considering piston + spring as system:

$$\Delta W_{gas} + \Delta W_{atm} = \Delta U_{piston} + \Delta U_{spring}$$

$$\Rightarrow \quad \Delta W_{gas} = mgL + \frac{1}{2}kL^2 + p_0AL$$

Now work done by piston on gas must be negative of work done by gas on piston (Kind of normal reaction pair), apart from this piston is also doing work against atmosphere and spring. (Also work done by piston on atmosphere is negative of work done by atmosphere on piston.

$$\therefore \quad \Delta W_{piston} = \Delta W_{pgas} + \Delta W_{pspring} + \Delta W_{patm}$$

$$\Rightarrow \Delta W_{piston} = -\left(mgL + \frac{1}{2}kL^2 + p_0AL\right) + \frac{1}{2}kL^2 + p_0AL$$

$$\Rightarrow \quad \Delta W_{piston} = -mgL \ ,$$

We could also write this directly by applying first law for piston alone.

$$\Delta Q = \Delta U + \Delta W = 0 \Rightarrow \Delta W = -\Delta U = -mgL \text{ Ans.}$$

Note: We could also find work done by directly integrating $dW = PdV$.

<u>WORK DONE IN AN ISOTHERMAL PROCESS.</u> Recall that for any polytropic process we proved that work done is given as:

$$\Delta W = n\Omega\Delta T = \frac{nR}{1-x}\Delta T$$ however this formula is useless for isothermal process since in isothermal process, $x = 1$ & $\Delta T = 0$, it yields a 0/0 form. However we can find the work here by direct integration:

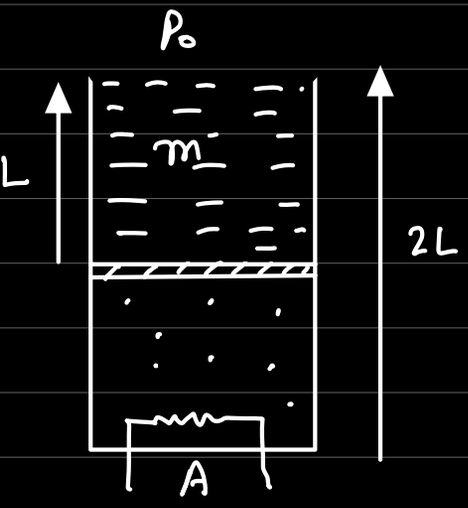
$$PV = nRT \Rightarrow dW = PdV = \frac{nRT}{V}dV \Rightarrow \Delta W = nRT\int_{V_i}^{V_f}(dV/V)$$

$$\Rightarrow \quad \boxed{\Delta W = nRT\ln\frac{V_f}{V_i} = nRT\ln\frac{P_i}{P_f} = P_iV_i\ln\frac{V_f}{V_i} = P_fV_f\ln\left(\frac{V_f}{V_i}\right)}$$

$\because nRT = P_iV_i = P_fV_f$

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

Example: A massless piston supported over a gas column in turn supports a water column as shown in the figure, now the system is supplied heat until water is expelled. The cylinder and piston are adiabatic. Find ΔU, ΔW and ΔQ for the gas for this process. Gas is monoatomic. What happens to temperature in process.



Sol: $\Delta U = \frac{f}{2}\Delta(PV) = \frac{3}{2}\left[P_0 \times 2AL - \left(p_0 + \frac{mg}{A}\right)AL\right]$

$\Rightarrow \Delta U = \frac{3}{2}AL\left[p_0 - \frac{mg}{A}\right]$

Now (considering piston + water as system)

$\Delta W_{gas} + \Delta W_{atm} = mg\frac{L}{2}$ (CM of water rises by $\frac{L}{2}$ if collected at brim).

$\Rightarrow \Delta W_{gas} = mg\frac{L}{2} + p_0 AL$

$\Rightarrow \Delta Q = \Delta U + \Delta W = \frac{5}{2}P_0 AL - mgL$

Note: Alternatively we could find work done by gas as $\Delta W = \int PdV$.

Here if $P_0 > mg/A$, then $\Delta U > 0$ (or temperature will increase)

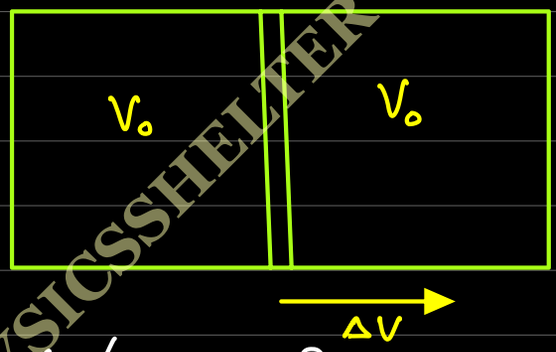
If $p_0 = mg/A$, final temperature is same as initial

(Still it is not isothermal process ∵ in between temperature may be something else).

YOUTUBE.COM/@THEPHYSICSSHELTER

Example: Irodov 2.36:

A piston can freely move inside a horizontal cylinder closed from both ends. Initially, the piston separates the inside space of the cylinder into two equal parts each of volume V_0, in which an ideal gas is contained under the same pressure p_0 and at the same temperature. What work has to be performed in order to increase isothermally the volume of one part of gas η times compared to that of the other by slowly moving the piston?



Since piston is massless, Net work done by external agents + Work done by gas on it must be zero

$\Rightarrow W_{ext} = -W_{gas}$. Let Volume swept by piston be ΔV

$$\Rightarrow W_{ext} = -\left[nRT \ln\left(\frac{V_0+\Delta V}{V_0}\right) + nRT \ln\left(\frac{V_0-\Delta V}{V_0}\right)\right]$$

$$= nRT \ln\left(\frac{V_0^2}{V_0^2-\Delta V^2}\right) \qquad \text{Now } \left(\frac{V_0+\Delta V}{V_0-\Delta V}\right) = \eta \Rightarrow \frac{\Delta V}{V_0} = \left(\frac{\eta-1}{\eta+1}\right)$$

$$\Rightarrow W_{ext} = p_0 V_0 \ln\left(\frac{V_0^2}{V_0^2\left[1-\left((\eta-1)/(\eta+1)\right)^2\right]}\right) = p_0 V_0 \ln\left(\frac{(\eta+1)^2}{4\eta}\right) \text{Ans.}$$

Molar Specific Heat Of A Polytropic Process : Recall that specific heat for any process is defined as $C = \frac{1}{n}\frac{dQ}{dT}$ (or heat required per mole per unit rise in temperature).

Now by first law, $dQ = dU + dW$

$$\Rightarrow C = \frac{1}{n}\frac{dU}{dT} + \frac{1}{n}\frac{dW}{dT} = C_V + \Omega = \frac{f}{2}R + \frac{R}{1-x}$$

$$\Rightarrow \boxed{C_x = \frac{f}{2}R + \frac{R}{1-x}}$$ ← First law for one mol per unit rise in Temp.

Now $\Delta Q = nC_x\Delta T$, $\Delta W = n\Omega\Delta T$, $\Delta U = nC_V\Delta T$

YOUTUBE.COM/@THEPHYSICSSHELTER

Also recall that $(f/2)R = R/\gamma-1 \Rightarrow$

$$\boxed{C_x = \frac{R}{\gamma-1} + \frac{R}{1-x}}$$

From this we can infer the ratio of $\Delta U/\Delta W$ or $\Delta U/\Delta Q$ or $\Delta W/\Delta Q$ for any polytropic process.

$$\frac{\Delta U}{\Delta W} = C_v/R = \left(\frac{1-x}{\gamma-1}\right) = \frac{f}{2}(1-x)$$

$$\Rightarrow \boxed{\frac{\Delta U}{\Delta W} = \frac{f}{2}(1-x)}$$

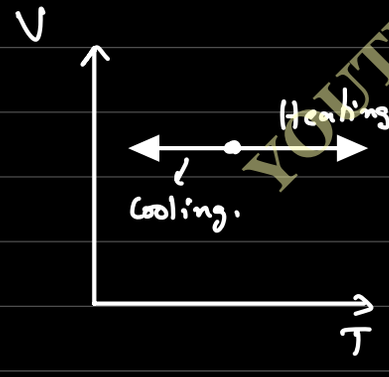
Molar Specific Heat Of A Mixture: If we have n_1 moles of gas with degree of freedom f_1 and n_2 mols of gas with degree f_2, and corresponding specific heats C_1 & C_2 for a particular polytropic coefficient then it can be easily shown that:

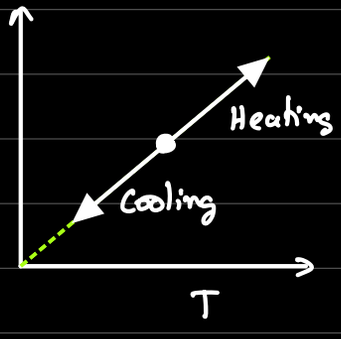
$$C_{eff} = \left(\frac{n_1C_1 + n_2C_2}{n_1+n_2}\right) \quad \left(\because U = \frac{n_1 f_1 RT}{2} + \frac{n_2 f_2 RT}{2}\right).$$

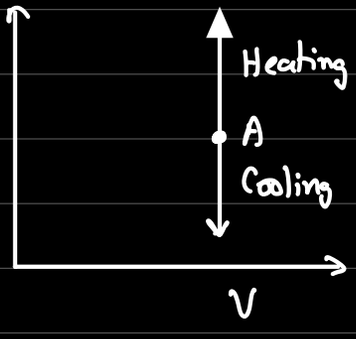
First Law Analyses For Special Polytropic Processes:

(I) Isochoric Process: $\boxed{\Delta W = 0}$ $(\because \Delta V = 0)$ $(x = \infty)$ $V = \text{Constt.}$

$$\Delta Q = \Delta U = nC_v\Delta T = n\frac{f}{2}R\Delta T = \frac{nR}{\gamma-1}\Delta T$$







(II) Isobaric Process: $P = \text{Constt}$, $x = 0$, $\Delta U = n\frac{f}{2}R\Delta T = \frac{nR}{\gamma-1}\Delta T$

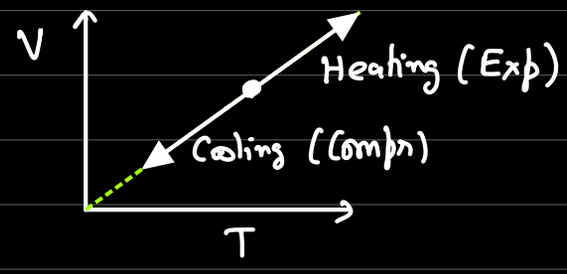
$$\boxed{\Delta W = nR\Delta T}$$

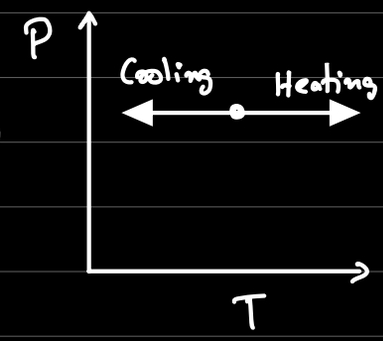
$$\Delta Q = n\left(\frac{f}{2}R + R\right)\Delta T = \left(\frac{f+2}{2}\right)nR\Delta T$$

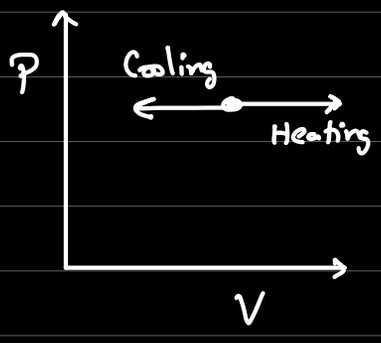
YOUTUBE.COM/@THEPHYSICSSHELTER

$\Rightarrow$ $\boxed{\Delta Q = \left(\frac{f+2}{2}\right) nR\Delta T}$

Plots:





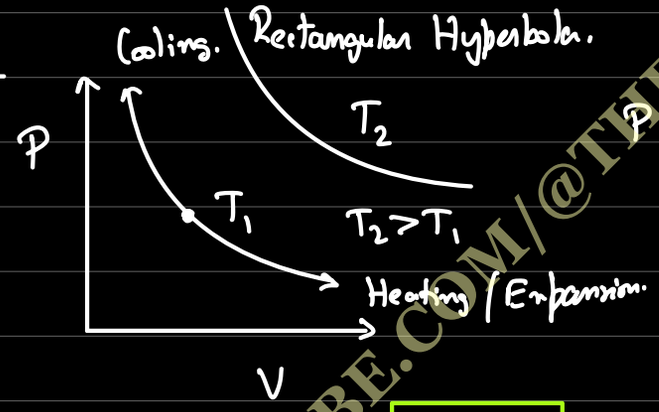


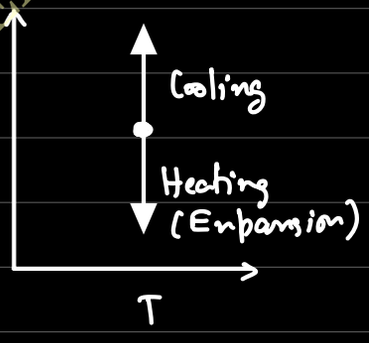
Gamma : $\gamma = C_p/C_v = 1+2/f$, $C_p = \gamma R/(\gamma-1)$ $C_v = R/(\gamma-1)$

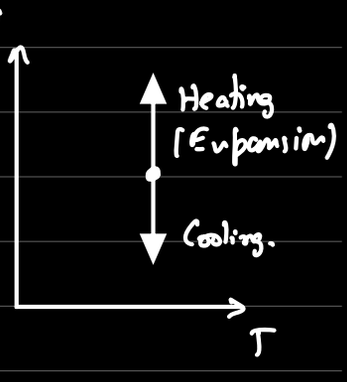
(III) Isothermal Process: $\boxed{\Delta U = 0}$.

$$\boxed{\Delta Q = \Delta W = nRT\ln\left(\frac{V_f}{V_i}\right) = nRT\ln\left(\frac{P_i}{P_f}\right) = PV\ln\left(\frac{V_f}{V_i}\right)}$$

Plots:







(IV) Adiabatic Process: $\boxed{\Delta Q = 0}$

Polytropic Coefficient : $\because$ No heat is supplied during Adiabatic Process

$\therefore$ Specific heat for adiabatic process must be zero

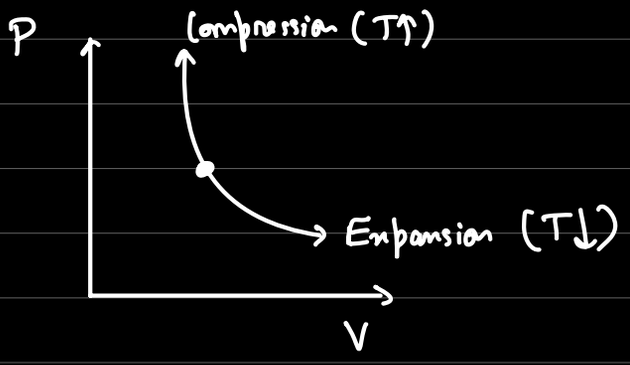
$\Rightarrow$ $C = \frac{R}{1-n} + \frac{R}{\gamma-1} = 0 \Rightarrow n = \gamma$ $\therefore$

For Adiabatic Process: $\boxed{PV^{\gamma} = C,\ TV^{\gamma-1} = C,\ P T^{\frac{\gamma}{1-\gamma}} = C}$

$$\boxed{\Delta U = n\frac{f}{2}R\Delta T = \frac{nR}{\gamma-1}\Delta T = \frac{P_f V_f - P_i V_i}{\gamma-1} = \frac{f}{2}(P_f V_f - P_i V_i) = -\Delta W}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

Plots: $PV^{\gamma} = \text{Const}$ $TV^{\gamma-1} = \text{Const}$ $PT^{\frac{\gamma}{1-\gamma}} = C$



To plot T-V Graph we need dT/dV and d^2T/dV^2

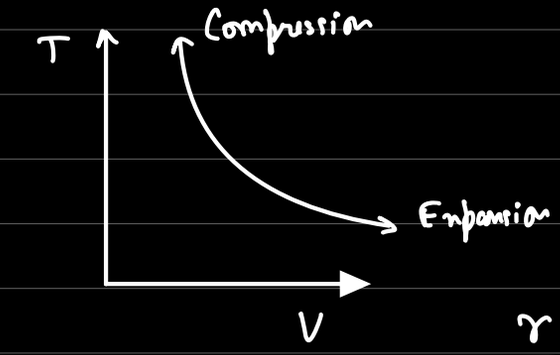
Now $\ln T + (\gamma-1)\ln V = \ln C$

$$\Rightarrow \frac{dT}{dV} = -(\gamma-1)\frac{T}{V} < 0$$

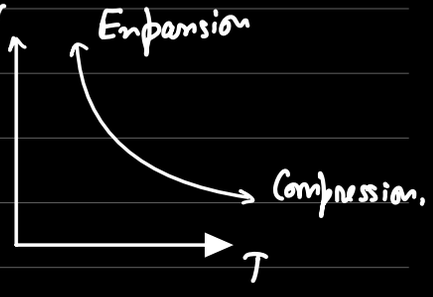
$$\frac{d^2T}{dV^2} = -(\gamma-1)\left[\frac{V\frac{dT}{dV} - T}{V^2}\right]$$

$$\Rightarrow \frac{d^2T}{dV^2} = -(\gamma-1)\left(-(\gamma-1)-1\right)\frac{T}{V^2} = \gamma(\gamma-1)\frac{T}{V^2} > 0$$

$\Rightarrow$ T-V Graph would look like:



For V-T Minor about 45° line



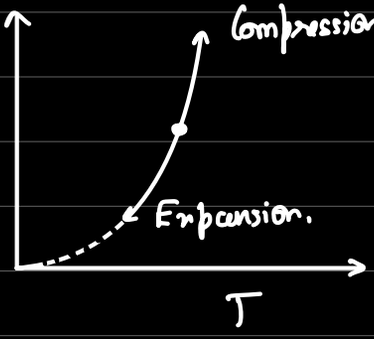
For PT Graph: $PT^{\frac{\gamma}{1-\gamma}} = C \Rightarrow \frac{dP}{dT} = -\left(\frac{\gamma}{1-\gamma}\right)\frac{P}{T} > 0$ $(2 > \gamma > 0)$

Now $\frac{d^2P}{dT^2} = \left(\frac{\gamma}{\gamma-1}\right)\left[\frac{T\frac{dP}{dT} - P}{T^2}\right] = \frac{\gamma P}{(\gamma-1)T^2}\left[\frac{\gamma}{\gamma-1} - 1\right]$

$$\Rightarrow \frac{d^2P}{dT^2} = \frac{\gamma}{(\gamma-1)^2}\frac{P}{T^2} > 0$$

eg. If $\gamma = 1.5 \Rightarrow PT^{\frac{1.5}{-0.5}} = C$

$\Rightarrow P \propto T^3$



Note: All polytropic Processes are monotonic (Don't have any maximum or minimum in between for P, T, V, ρ, Q, U or W, maximum and minimum will always occur at one of the end points of the process.

→ The technique use for plotting P-T graph above can be used for figuring out shape of any polytropic Process (e.g. P-ρ graph)

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

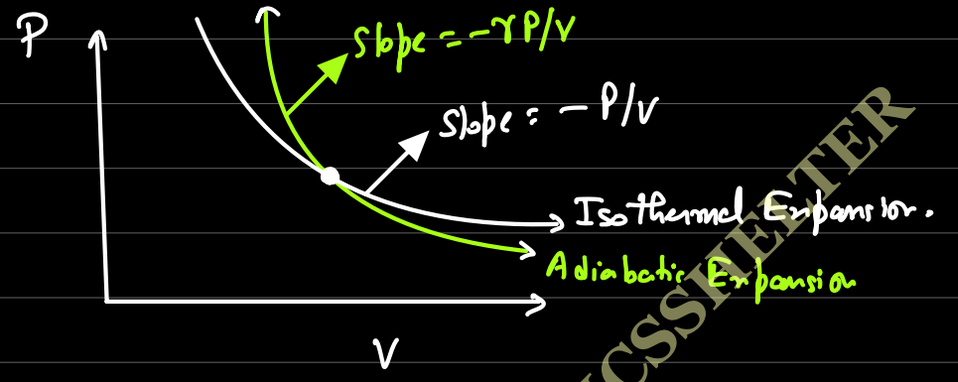
For P-ρ graph $\frac{dP}{d\rho}$ will tell us rise or fall and $\frac{d^2P}{d\rho^2}$ will tell Curvature.

Comparison Of Slope Of Isothermal & Adiabatic Lines on P-V Diagram.

For a polytropic process slope of PV curve is given by $\frac{dP}{dV} = -\frac{xP}{V}$

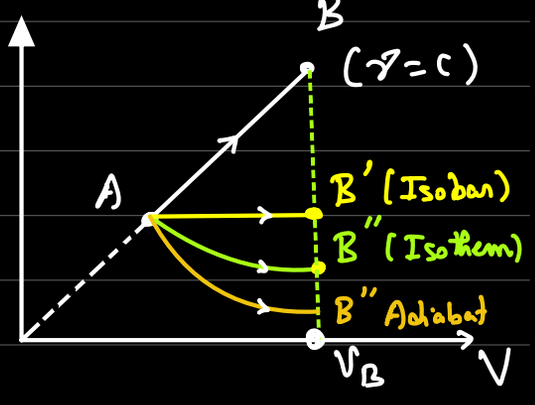
For isothermal curve, $x = 1$ while for adiabatic $x = \gamma$.

Thus at any particular state of gas, Adiabatic Curve will always be steeper than isothermal curve.



Example: From a certain state a gas is to be expanded to a certain volume either through isobar, isotherm, adiabat or constant collision frequency process. Arrange the processes in decreasing order of work done.

Solution: The area under the P-V diagram represents the work done in any process. The four polytropes are as shown in adjacent figure. Clearly maximum area is under $\gamma = C$ followed by $P = C$, $T = C$ & $\Delta Q = 0$ lines.



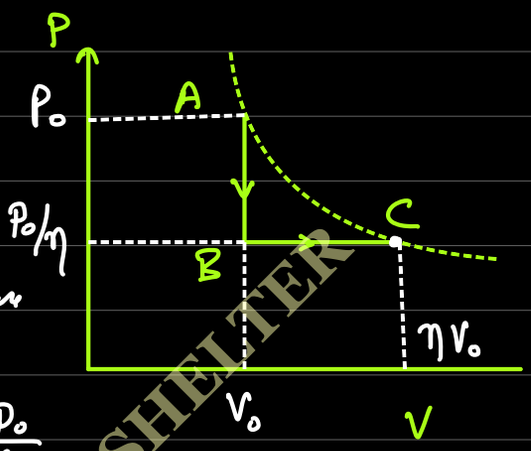
Thus:

$$\Delta W_\gamma > \Delta W_P > \Delta W_T > \Delta W_Q \quad \text{Ans.}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

Example: Irodov 2.32

Two moles of a certain ideal gas at a temperature $T_0 = 300$ K were cooled isochorically so that the gas pressure reduced $\eta = 2.0$ times. Then, as a result of the isobaric process, the gas expanded till its temperature got back to the initial value. Find the total amount of heat absorbed by the gas in this process.



Solution: Along isotherm $A \to C$, $PV =$ Constt.

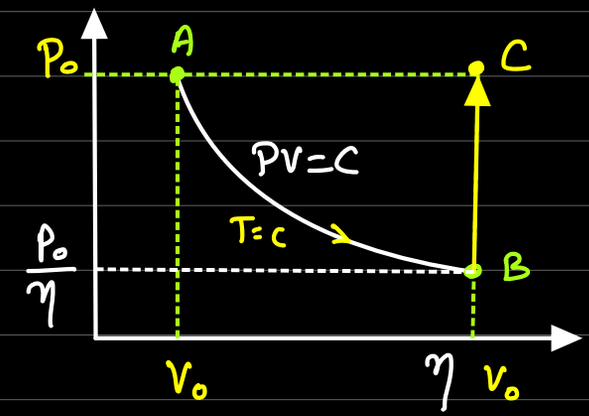
$\Rightarrow P_0 V_0 = \frac{P_0}{\eta} \times V_C \Rightarrow V_C = \eta V_0$

Now for Process ABC, $\Delta U = 0$ ∵ temperature change is zero. ∴ $\Delta Q = \Delta W =$ Area under the graph. $\Rightarrow \Delta W = (\eta - 1) V_0 \times \frac{P_0}{\eta}$

$$\Rightarrow \Delta W = P_0 V_0 \left(\frac{\eta - 1}{\eta}\right) = nRT_0 \left(\frac{\eta - 1}{\eta}\right) = \Delta Q$$

Example: Irodov 2.37

Three moles of an ideal gas being initially at a temperature $T_0 = 273$ K were isothermally expanded $n = 5.0$ times its initial volume and then isochorically heated so that the pressure in the final state became equal to that in the initial state. The total amount of heat transferred to the gas during the process equals $Q = 80$ kJ. Find the ratio $\gamma = C_p/C_V$ for this gas.



Solution: For $A \to B$ $\Delta U = 0$,

$\Delta Q = \Delta W = nRT \ln(V_f/V_i)$

$\Rightarrow \Delta W_{AB} = nRT_0 \ln(\eta) = \Delta Q_{AB}$

For $B \to C$, $\Delta Q = \Delta U = \frac{f}{2} \Delta(PV)$

$$\Rightarrow \Delta Q = \frac{f}{2}\left[P_0 \times \eta V_0 - \frac{P_0}{\eta} \times \eta V_0\right)$$

$$\Rightarrow \Delta Q_{BC} = \frac{f}{2}(\eta - 1) P_0 V_0 = \frac{\eta f}{2} R T_0 (\eta - 1)$$

$$\Rightarrow \Delta Q = \Delta Q_{AB} + \Delta Q_{BC} = nRT_0 \left[\frac{f}{2}(\eta - 1) + \ln \eta\right] = Q, \quad \gamma = 1 + \frac{2}{f}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

$$\Rightarrow \frac{f}{2} = \left(\frac{Q}{nRT_0} - \ln\eta\right)\frac{1}{\eta-1} \quad \Rightarrow \gamma = 1 + \frac{(\eta-1)}{\left(\frac{Q}{nRT_0} - \ln\eta\right)} \text{ Ans.}$$

EXAMPLE: IRODOV: 2·49:

An ideal gas whose adiabatic exponent equals γ is expanded so that the amount of heat transferred to the gas is equal to the decrease of its internal energy. Find:

(a) the molar heat capacity of the gas in this process;

(b) the equation of the process in the variables T, V;

(c) the work performed by one mole of the gas when its volume increases η times if the initial temperature of the gas is T_0.

Solution: $\Delta Q = -\Delta U$ Also $\Delta Q \propto \frac{R}{\gamma-1} + \frac{R}{1-x}$, $\Delta U \propto \frac{R}{\gamma-1}$

$$\Rightarrow \frac{R}{\gamma-1} + \frac{R}{1-x} = -\left(\frac{R}{\gamma-1}\right) \Rightarrow \frac{R}{1-x} = \frac{-2R}{\gamma-1}$$

$$\Rightarrow C = \frac{R}{\gamma-1} + \frac{R}{1-x} = \frac{-R}{\gamma-1} \text{ Ans., Also } x = (1+\gamma)/2.$$

Now Procm: $TV^{x-1} = C \Rightarrow TV^{(\gamma-1)/2} = C$ Ans.

Now $\Delta W = \frac{nR}{1-x}\Delta T = \frac{-2nR}{\gamma-1}\Delta T$

Now $T_0 V_i^{(\gamma-1)/2} = T_f(\eta V_i)^{\gamma-1/2} \Rightarrow T_f = T_0\left(\frac{1}{\eta}\right)^{(\gamma-1)/2}$

$$\Rightarrow \Delta W = \frac{-2nRT_0}{\gamma-1}\left[\left(\frac{1}{\eta}\right)^{(\gamma-1)/2} - 1\right] = \frac{2nRT_0}{\gamma-1}\left[1 - \eta^{(1-\gamma)/2}\right] \text{ Ans.}$$

EXAMPLE: IRODOV: 2·53

One mole of an ideal gas whose adiabatic exponent equals γ undergoes a process $p = p_0 + \alpha/V$, where p_0 and α are positive constants. Find:

(a) heat capacity of the gas as a function of its volume;

(b) the internal energy increment of the gas, the work performed by it, and the amount of heat transferred to the gas, if its volume increased from V_1 to V_2.

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: $P = P_0 + \alpha/V$ ① $PV = RT$ ② (Considering 1 mol of gas)

From ① & ② :

$$\left(P_0 + \frac{\alpha}{V}\right)V = RT \Rightarrow P_0V + \alpha = RT \Rightarrow P_0\frac{dV}{dT} = R$$

$$\Rightarrow \frac{P\,dV}{dT} = \frac{PR}{P_0} = \frac{1}{P_0}\left(P_0 + \frac{\alpha}{V}\right)R = R + \frac{\alpha R}{P_0V} = \left.\frac{dW}{dT}\right|_{1\,mol} = \Omega$$

$$\Rightarrow C = C_v + \left.\frac{dW}{dT}\right|_{1\,mol} = C_v + R\left[1 + \frac{\alpha}{P_0V}\right] = (C_v + R) + \frac{\alpha R}{P_0V} = \frac{\gamma R}{\gamma - 1} + \frac{\alpha R}{P_0V}$$

Now $\Delta U = \frac{f}{2}\Delta(PV) = \frac{1}{\gamma - 1}\Delta(PV) = \frac{1}{\gamma - 1}\left[V_2\left[P_0 + \frac{\alpha}{V_2}\right] - V_1\left[P_0 + \frac{\alpha}{V_1}\right]\right]$

$$\Rightarrow \Delta U = \frac{P_0}{\gamma - 1}\left[V_2 - V_1\right] \text{ ans.}$$

Now $\Delta W = \int P\,dV = \int_{V_1}^{V_2}\left(P_0 + \frac{\alpha}{V}\right)dV = P_0(V_2 - V_1) + \alpha \ln\left(\frac{V_2}{V_1}\right)$

$$\Rightarrow \Delta Q = \Delta U + \Delta W = \frac{\gamma P_0}{\gamma - 1}(V_2 - V_1) + \alpha \ln\left(\frac{V_2}{V_1}\right) \text{ ans.}$$

EXAMPLE : IRODOV 2·56

An ideal gas has an adiabatic exponent γ. In some process its molar heat capacity varies as $C = \alpha/T$, where α is a constant, Find :

(a) the work performed by one mole of the gas during its heating from the temperature T_0 to the temperature η times higher;

(b) the equation of the process in the variables p, V.

Solution: $\Delta Q = \int C\,dT = \int_{T_0}^{\eta T_0}\frac{\alpha}{T}\,dT = \alpha \ln \eta$

Now $\Delta U = n C_v \Delta T = C_v(\eta - 1)T_0 \Rightarrow \Delta W = \Delta Q - \Delta U \Rightarrow$

$$\Delta W = \alpha \ln(\eta) - C_v(\eta - 1)T_0 = \alpha \ln \eta - \frac{R}{\gamma - 1}(\eta - 1)T_0$$

Now $\frac{P\,dV}{dT} = \frac{\alpha}{T} - \frac{R}{\gamma - 1}$ ① and $PV = RT$ ② ($\because \frac{dW}{dT} = C - C_v$)

$$\Rightarrow \frac{RT}{V}\frac{dV}{dT} = \frac{\alpha}{T} - \frac{R}{\gamma - 1} \Rightarrow (\gamma - 1)\frac{dV}{V} = \frac{\alpha(\gamma - 1)\,dT}{RT^2} - \frac{dT}{T}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

$$\Rightarrow (\gamma-1)\ln V = \frac{-\alpha(\gamma-1)}{RT} - \ln T + C$$

$$\Rightarrow T V^{\gamma-1} e^{\frac{-\alpha(\gamma-1)}{RT}} = C \quad \Rightarrow P V^{\gamma} e^{\frac{-\alpha(\gamma-1)}{RT}} = C \quad (\because TV^{\gamma-1} \equiv PV^{\gamma})$$

$$\Rightarrow \text{Required Process } P V^{\gamma} e^{\frac{-\alpha(\gamma-1)}{PV}} = C \quad \text{Ans.}$$

Maximum / Minimum Temperature In Non-Polytropic Processes:

While for polytropic processes maximum or minimum temperature always occurs at end points of the process. For non-polytropic processes it might occur somewhere in between. It can be shown that at points of maximum or minimum temperature occuring in between a process, the slope of the process must be same as the slope of the isotherm on the diagram. i.e. $\frac{\partial P}{\partial V} = -P/V$.

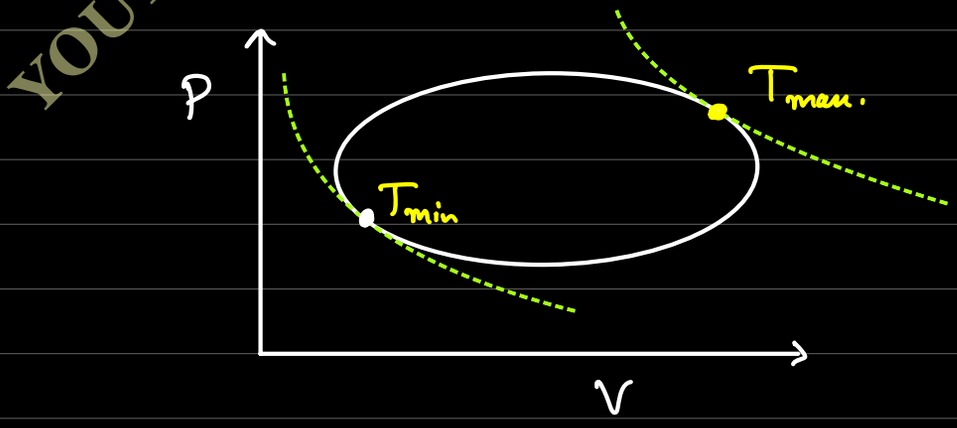
Proof: Let $P = f(V)$, also $PV = RT$,

$$\Rightarrow f(V)\,V = RT \Rightarrow \ln R + \ln T = \ln f(V) + \ln(V)$$

Differentiating wrt V, $\frac{1}{T}\frac{dT}{dV} = \frac{1}{f(V)} \cdot f'(V) + \frac{1}{V} = 0$

$(\because$ for max / min T, $\partial T/\partial V = 0) \Rightarrow f'(V) = -\frac{f(V)}{V} = -\frac{P}{V}$

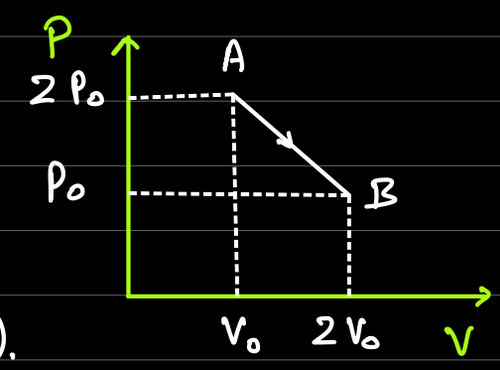
Thus slope of the process is same as that of the isotherm or we can also say that isotherm is the tangent to the curve.



We could also see this directly from graph, As we move away from the origin, the isotherms have higher and higher temperatures, so we are looking for a point on our curve which also lies on farthest isotherm ∴ it must be a tangent.

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: A certain process is shown from A to B, locate the point of maximum temperature. Also find max temperature considering n moles.



Solution: $\frac{P}{P_0} + \frac{V}{V_0} = 3$ ① (Curve equation).

$\Rightarrow \frac{dP}{dV} = -P_0/V_0 \Rightarrow$ At point of max temp, $-\frac{P}{V} = -\frac{P_0}{V_0}$ ②

($\because dP/dV = -P/V$ for isotherm)

Solving ① & ③, $P = \frac{3}{2}P_0 \quad V = \frac{3}{2}V_0 \Rightarrow T_{max} = \frac{PV}{nR} = \frac{9}{4}\frac{P_0V_0}{nR}$ Ans.

LOCATING HEAT REVERSAL POINT IN NON POLYTROPIC PROCESSES: In polytropic processes the heat flow direction does not change as the volume is increased however in non polytropic process, it might be possible that the system absorbs heat till a certain point and after that it may start rejecting heat (or other way round). For purpose of calculating cycle efficiencies with non-polytropic process it may be necessary to locate the heat reversal point. It can be shown that at heat reversal point, slope of the process is equal to the slope of the adiabat i.e $\frac{dP}{dV} = -\frac{\gamma P}{V}$.

Proof: At heat reversal point, $\frac{dQ}{dV} = 0$, $dQ = dU + PdV$

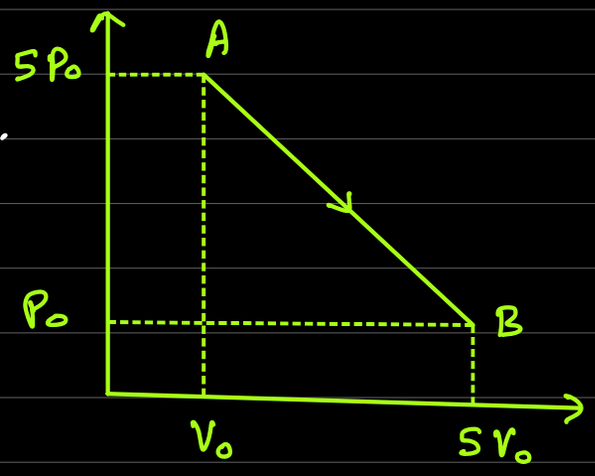
$\Rightarrow \frac{dU}{dV} + P = 0$, Now $U = \frac{f}{2}PV \Rightarrow \frac{dU}{dV} = \frac{f}{2}\left[P + V\frac{dP}{dV}\right]$

$\Rightarrow \left(1+\frac{f}{2}\right)P + \frac{f}{2}V\frac{dP}{dV} = 0 \Rightarrow \frac{dP}{dV} = -\left(1+\frac{2}{f}\right)\frac{P}{V} = -\frac{\gamma P}{V}$.

$\Rightarrow \frac{dP}{dV} = -\frac{\gamma P}{V}$ Or we can say that slope of the process is equal to the slope of the adiabat (or adiabat is the tangent to the curve).

YOUTUBE.COM/@THEPHYSICSSHELTER

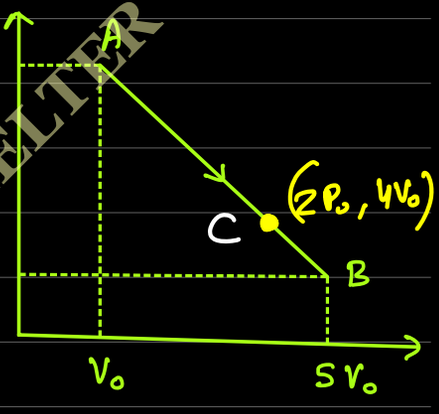
EXAMPLE: A gas having $\gamma = 2$ undergoes a process $A \to B$ as shown in the figure. Locate the point of heat reversal. Also find the heat supplied from A to this point.



Solution: $\frac{P}{P_0} + \frac{V}{V_0} = 6$ ①

$\Rightarrow \frac{dP}{dV} = -\frac{P_0}{V_0} = -2\frac{P}{V}$ ② ($\because$ for heat reversal $\frac{dP}{dV} = -\gamma P/V$)

Solving ① & ② : $P = 2p_0$, $V = 4V_0$.



Now for heat supplied $A \to C$: $\Delta Q = \Delta U + \Delta W$

Now: $\Delta U = \frac{f}{2}\Delta(PV) = \frac{1}{\gamma - 1}\Delta(PV) = \Delta(PV)$ ($\because \gamma = 2$)

$\Rightarrow \Delta U = (2P_0 \times 4V_0 - 5P_0 \times V_0) = 3P_0V_0$

ΔW = Area under the curve from $A \to C$ $\Rightarrow$

$\Delta W = \frac{1}{2}[5P_0 + 2P_0] \times 3V_0 = \frac{21}{2}P_0V_0$ $\Rightarrow \Delta Q = \Delta U + \Delta W = \frac{27}{2}P_0V_0$

$\Rightarrow \Delta Q = \frac{27}{2}P_0V_0$ Ans.

Q. What is ΔQ for $C \to B$ for the situation of previous example?

Solution: $\Delta Q = \Delta U + \Delta W = \frac{\Delta(PV)}{\gamma - 1}$ + Area under the curve.

Now $\Delta U = P_0 \times 5V_0 - (2P_0 \times 4V_0) = -3P_0V_0$

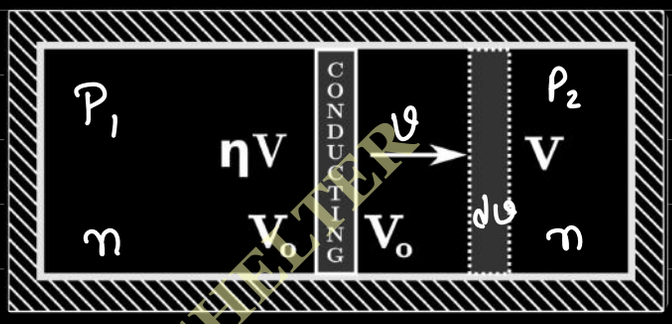
$\Delta W = \frac{1}{2}(2P_0 + P_0) \times (5V_0 - 4V_0) = \frac{3}{2}P_0V_0 \Rightarrow \Delta Q = \Delta U + \Delta W$

$\Rightarrow \Delta Q = -\frac{3}{2}P_0V_0$ Ans. [System absorbs heat from $A \to C$ and rejects heat from $C \to B$].

YOUTUBE.COM/@THEPHYSICSSHELTER

Example: Irodov 2·41:

A heat-conducting piston can freely move inside a closed thermally insulated cylinder with an ideal gas. In equilibrium the piston divides the cylinder into two equal parts, the gas temperature being equal to T_0. The piston is slowly displaced. Find the gas temperature as a function of the ratio η of the volumes of the greater and smaller sections. The adiabatic exponent of the gas is equal to γ.



Solution: Let volume swept by piston at any time be v and let dv be additional swept volume. Let temperature at this instant be T.

$\Rightarrow P_1 = \eta RT/(V_0+v)$ ① $P_2 = \eta RT/(V_0-v)$ ②

Now since no heat is supplied to the system, by first law:

$dQ = dU + dW \Rightarrow dW = -dU$ ($\because dQ = 0$ for system as a whole).

$\Rightarrow (P_1 - P_2)\,dv = -2\eta \frac{f}{2} R\,dT$ ③, Now using ① & ② in ③:

$$\eta RT\left[\frac{1}{V_0+v} - \frac{1}{V_0-v}\right]dv = -2\eta\frac{f}{2}R\,dT \Rightarrow \frac{v\,dv}{V_0^2 - v^2} = \frac{f}{2}\frac{dT}{T} \quad ④$$

Now final volume ratio: $\frac{V_0 + v_{final}}{V_0 - v_{final}} = \eta \Rightarrow v_{final} = V_0\left(\frac{\eta-1}{\eta+1}\right)$

Now Integrating ④ in appropriate limits:

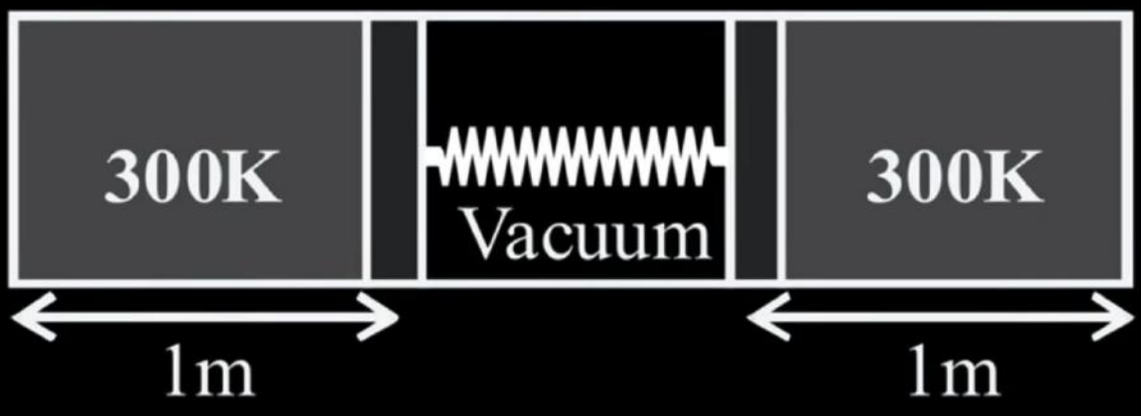
$$-\frac{1}{2}\left[\ln(V_0^2 - v^2)\right]_0^{v_{final}} = \frac{f}{2}\ln(T/T_0) \Rightarrow T = T_0\left[\frac{V_0^2}{V_0^2 - v_{final}^2}\right]^{\frac{1}{f}}$$

Now $\gamma = 1 + 2/f \Rightarrow f = 2/(\gamma-1)$, Putting v_{final} in T:

$$T = T_0\left[\frac{(\eta+1)^2}{4\eta}\right]^{\frac{\gamma-1}{2}} \text{ Ans.}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE:

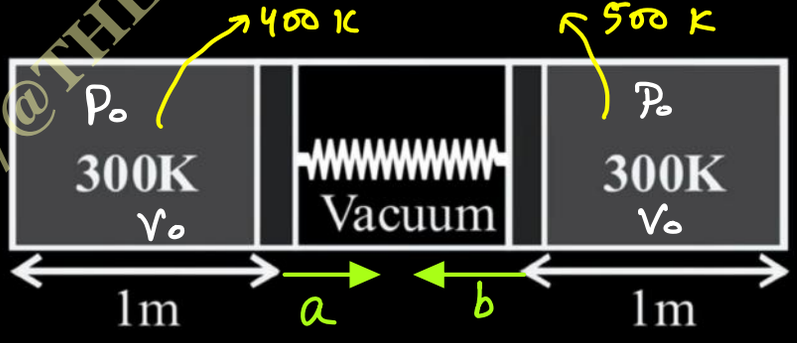


Consider the shown diagram where the two chambers separated by piston-spring arrangement contain equal amounts of certain ideal gas. Initially when the temperatures of the gas in both the chambers are kept at 300 K, the compression in the spring is 1 m. The temperature of the left and the right chambers are now raised to 400 K and 500 K respectively. If the pistons are free to slide, the final compression in the spring (in cm) is 130 times x. Find value of x. [take $\sqrt{13} = 3.6$]

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: The pressures in chambers are directly proportional to compression in the spring.

Let the pistons move through distances a & b as shown in the figure.



Also volumes are directly proportional to the lengths of chambers.

Now $n = \frac{PV}{RT}$, since number of moles in each chamber is conserved we can write: $\frac{P_0 V_0}{R \times 300} = \frac{P_0 (1+a+b) V_0}{1} \left(\frac{1+a}{1}\right) \times \frac{1}{400 R}$

or $(1+a+b)(1+a) = 4/3$ ①, Similarly for the other chamber:

$(1+a+b)(1+b) = 5/3$ ②

$\Rightarrow$ $(1+a+b)(2+a+b) = 9/3 = 3$ (Adding ① & ②)

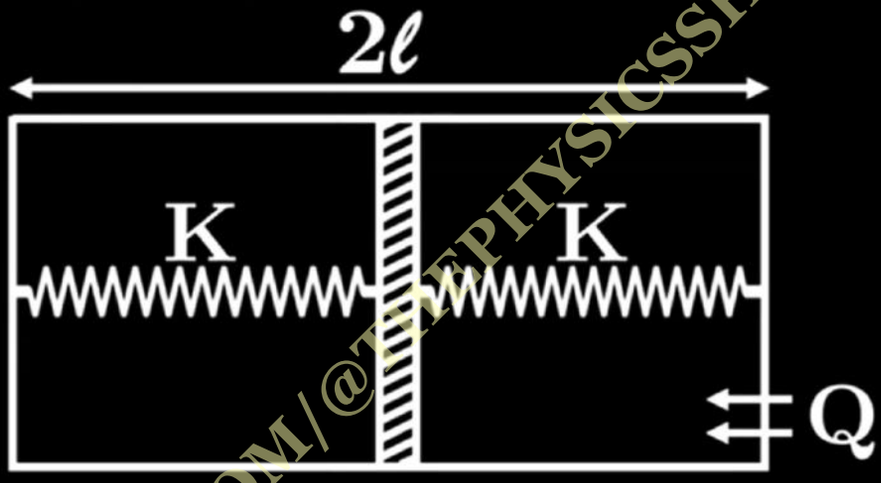
$\Rightarrow$ $y^2 + 3y - 1 = 0$ where $y = a+b$.

$\Rightarrow \quad y = \frac{-3+\sqrt{13}}{2} \quad \Rightarrow$ final compression $= 1+y = \frac{\sqrt{13}-1}{2} = 1.3x$

$\Rightarrow \quad x = 1$ Ans.

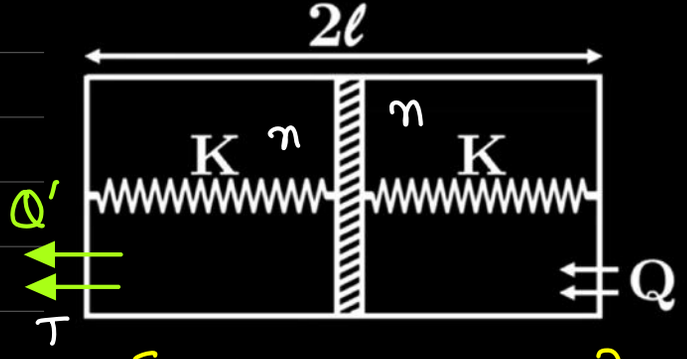
EXAMPLE: KROTOV: 2.12

2.12*. A horizontal cylindrical vessel of length $2l$ is separated by a thin heat insulating piston into two equal parts each of which contains n moles of an ideal monatomic gas at a temperature T. The piston is connected to the end faces of the vessel by undeformed springs of rigidity K each (Fig). When an amount of heat Q is supplied to the gas in the right part, the piston is displaced to the left by a distance $x = l/2$.
Determine the amount of heat Q' given away at the temperature T to a thermostat with which the gas in the left part is in thermal contact all the time.



YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: Let the system reject a heat Q' to the thermostat. Here we define system to be cylinder including all its contents (i.e springs, piston and gas).
Now ∵ system is not pushing any boundary interface between the system and it surrounding ∴ $\Delta W_{sys} = 0$
$\Rightarrow \Delta Q_{net} = \Delta U_{sys} \Rightarrow Q - Q' = \Delta U_{sys}$.



[Volume Swept = $V_0/2$]

Now for right chamber, $\Delta U_R = \frac{f}{2}\Delta(PV)$. Now pressure in right chamber increases due to spring force changes and pressure change in left side.

Now left chamber undergoes isothermal compression. $\Rightarrow P_f V_f = P_i V_i$

$$\Rightarrow P_L = \frac{P_0 V_0}{(V_0/2)} = 2P_0 \Rightarrow P_R = 2P_0 + \left(\frac{kl}{A}\right) = 2P_0 + \frac{k l^2}{V_0}$$

$$\Rightarrow \Delta U_R = \frac{3}{2}\left[\left(2P_0 + \frac{k l^2}{V_0}\right)\left(\frac{3V_0}{2}\right) - P_0 V_0\right] = \frac{9}{4} k l^2 + 3P_0 V_0$$

$$\Rightarrow \Delta U_R = \frac{9}{4} k l^2 + 3nRT$$

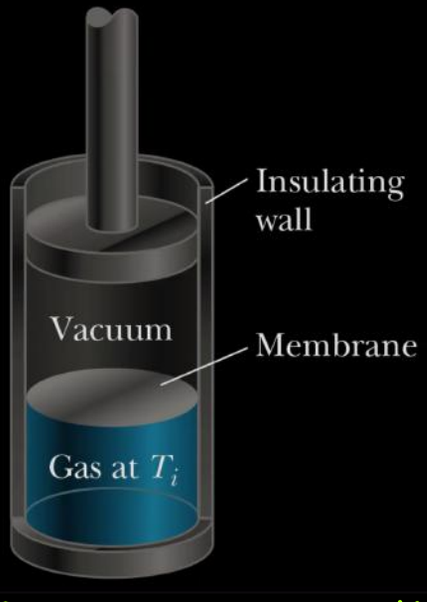
Now for left chamber gas, $\Delta T = 0 \Rightarrow \Delta U_L = 0$

For springs, $\Delta U_{sp} = \frac{1}{2} k \left(\frac{l}{2}\right)^2 \times 2 = \frac{1}{4} k l^2$

$$\Rightarrow \text{Net } \Delta U_{sys} = \Delta U_R + \Delta U_L + \Delta U_{sp} = 3nRT + \frac{5}{2} k l^2$$

Now $Q' = Q - \Delta U_{sys} = Q - 3nRT - \frac{5}{2} k l^2$ Ans.

FREE EXPANSION PROCESS: (IRREVERSIBLE): A process in which it is not possible to define the state of a gas in terms of a unique pressure, volume and temperature can't be represented on a graph. For example if a gas is suddenly allowed to expand, (For example by removing suddenly a weight from the piston). One such irreversible process is free expansion process in which a well insulated vessel is partitioned in to two chambers with gas in one partition and vacuum in the other part. Then the partition is removed to allow adiabatic free expansion of the Gas.



In such a process ∵ walls are adiabatic
∴ $\Delta Q = 0$, Also since vessel is not expanding against any external boundary
∴ $\Delta W = 0$ and by first law then:
$\Delta U = 0$ ($\because \Delta Q = \Delta W = 0$).

Note: That Free expansion is simultaneously adiabatic and isothermal.
→ It satisfies $PV = C$ and not $PV^\gamma = C$ ← Only valid for reversible

YOUTUBE.COM/@THEPHYSICSSHELTER

→ Caution: Never Use $PV^\gamma = C$ in any irreversible Adiabatic Process (Sudden removal of weights from piston). In such cases, work done by gas can be indirectly found by applying work energy theorem on piston and if external pressure is constant then work done by external pressure on piston is $W_{ext} = -p_0 \Delta V$.

EXAMPLE: (RESNICK HALLIDAY WALKER (GUPTA)):

Cylinder with massive piston

Consider n moles of an ideal monoatomic gas placed in a vertical cylinder. The top of the cylinder is closed by a piston of mass M and cross section A (Fig. 21-20a). Initially the piston is fixed by external stops, and the gas has volume V_0 and temperature T_0. Next, the piston is released by removing stops, and after several oscillations comes to a stop. Disregarding friction and the heat capacity of the piston and cylinder, find the temperature and volume of the gas at equilibrium. The system is thermally isolated, and the pressure outside the cylinder is p_a.



YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: Let final volume of gas be V. $\Rightarrow$ Height fallen $h = (V_0 - V)/A$.

Final Pressure, $P = \frac{Mg}{A} + P_a$, Now $\because \Delta Q = 0 \Rightarrow \Delta U = -\Delta W$

Considering piston + Gas as the system:

$$P_a(V_0 - V) = -Mg(V_0 - V)/A + \frac{3}{2}\left[\left(P_a + \frac{Mg}{A}\right) \times V - nRT_0\right] \quad (1)$$

$(\because \Delta U = \frac{f}{2} PV - \frac{nf}{2} RT_0)$.

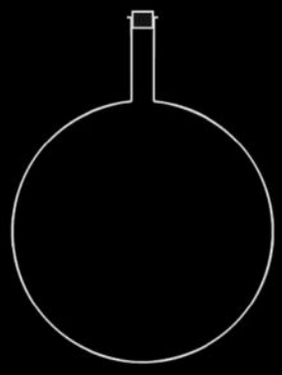
Solving (1) for V, we get: $V = \frac{2V_0}{5} + \frac{3}{5}\frac{nRT_0}{(P_a + Mg/A)}$ Ans.

Now final temperature: $T = \left(\frac{P_f V_f}{nR}\right) = \frac{(P_a + Mg/A) \times V}{nR}$

$\Rightarrow T = \frac{3T_0}{5} + \frac{2}{5}\frac{(P_a + Mg/A)}{5nR}$ Ans.

Example: Pathfinder MCQ 22:

A heat-insulated, rigid flask is evacuated and its opening is sealed with a cork. The flask is held at rest in a large chamber filled with an ideal mono-atomic gas at temperature 27 °C. The cork is suddenly removed and air quickly fills the flask. If volume of the flask is negligible as compared to the volume of the chamber, what is the temperature of the air inside the flask immediately after a thermodynamic equilibrium is attained in the flask?



(a) –93 °C (b) 27 °C
(c) 147 °C (d) 227 °C

Solution: Imagine a "cloud" of monoatomic gas filling the vessel. Here our system is cloud + the vessel volume. $\Delta Q = 0 \therefore \Delta W = -\Delta U$,
Let chamber pressure be P_0, then final pressure in vessel must also be P_0. Let volume of flask be V_0 and that of cloud be V

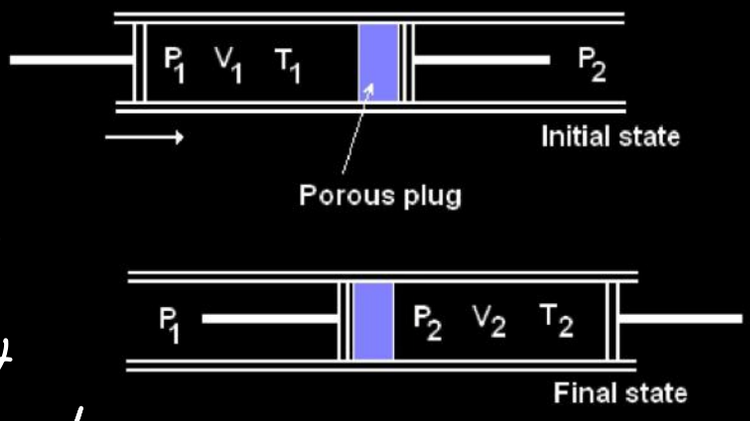


$$\Rightarrow \quad P_0 V = \frac{3}{2} nR(T - T_0) \Rightarrow nRT_0 = \frac{3}{2} nR(T - T_0)$$

$$\Rightarrow \quad T = \frac{5}{3} T_0 = \left[\frac{5}{3} \times 300 - 273\right] °C = 227°C \text{ Ans.}$$

Isenthalpic Throttling (Joule Thomson Effect):

Imagine a gas at pressure P_1 being pushed through a porous plug to push another piston on which a constant pressure P_2 is acting. The whole apparatus is insulated. It can be shown that in such a process enthalpy is conserved (i.e $U + PV =$ constant).



Proof: $\because \Delta Q = 0 \therefore \Delta W = -\Delta U \Rightarrow (P_1 V_1 - P_2 V_2) = (U_2 - U_1)$

$\Rightarrow U_1 + P_1 V_1 = U_2 + P_2 V_2$ or $H_1 = H_2$ (Isenthalpic Process).

YOUTUBE.COM/@THEPHYSICSSHELTER

Note that for an ideal gas temperature will not change in isenthalpic process. This can be shown as follows:

$$U + PV = \text{Constt} \implies n\frac{f}{2}RT + nRT = \text{Constt}$$

$$\implies T = \text{Constt}.$$

However for Real gases, heating or cooling can be achieved through Joule Thomson throttling process. (Recall that for a Van Der Waal Gas, $U = \frac{f}{2}RT - \frac{a}{V_m}$).

YOUTUBE.COM/@THEPHYSICSSHELTER